

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
Department of Computer Science and Engineering
Assignment 1

Academic Year: 2023-2024 EVEN

Batch: 2021-2025

III Year - VI Semester

Course Code	UCS2604	Course Name	Principles of Machine Learning					
Course Type	Theory	Course Category	Professional Core (PC)	L	T	P	E	C
				3	0	0	0	3
Regulation	2021		Academic Year	2023-2024 Even				
Degree and Branch	B.E. Computer Science & Engineering		Batch	2021-2025				
Semester		VI	Faculty Name	Dr. Y. V. Lokeswari & Ms. Nilu R. Salim				
Department Offering the Course			Computer Science and Engineering					

COURSE OUTCOMES

Upon completion of this course, the students should be able to:

1. Explain the basic concepts of machine learning (K2)
 2. Apply supervised algorithms for different classification problems (K3)
 3. Explain the need for ensemble methods (K2)
 4. Apply unsupervised and reinforcement learning techniques to various problems (K3)
 5. Apply dimensionality reduction and optimization techniques (K3)
-
1. Consider the given MNIST dataset and apply Implement the Multi-Layer Perceptron to classify the Handwritten characters. [CO2, K3] [1.1.1, 1.4.1, 2.3.1]

MNIST is a collection of handwritten digits ranging from the number 0 to 9.

It has a training set of 60,000 images, and 10,000 test images that are classified into corresponding categories or labels.

<https://www.kaggle.com/code/prashant111/mnist-deep-neural-network-with-keras/input>

2. Construct a KD-Tree for the given pair of data points. (7,8), (12,3), (14,1), (4,12), (9,1), (2,7), and (10,19).
Identify the cutting dimension and perform space partition.
Search for the minimum element in the KD- Tree.
Check the nearest neighbour for the query point (11, 5). [CO3, K3] [1.1.1, 13.3.1]
Refer to <https://www.cs.cmu.edu/~ckingsf/bioinfo-lectures/kdtrees.pdf>
<https://www.geeksforgeeks.org/search-and-insertion-in-k-dimensional-tree/>

Faculty

PAC

HoD-CSE

Y. V. Lokeswari
Nilu R. Salim

Dr. T. T. Mirnalinee